GEOGRAPHY

PHYSICAL GEOGRAPHY

MODULE -3

- FEATURES OF SUN
- PHOTOSPHERE, CHROMOSPHERE, CORONA
- ASTEROIDS, DWARF PLANETS
- COMETS, METEOROIDS

CELESTIAL BODY

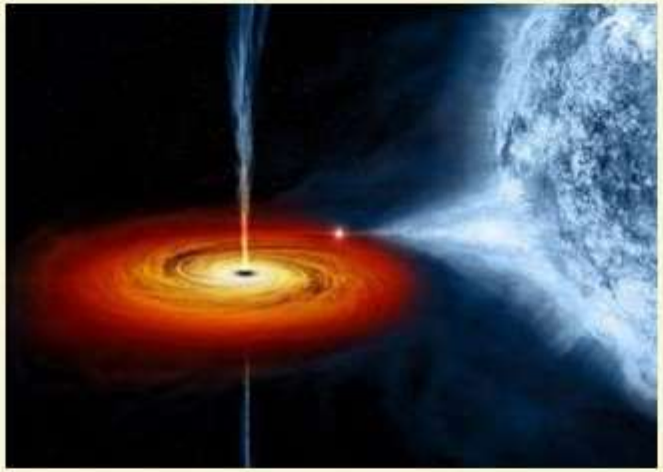
- Any natural body outside of the Earth's atmosphere, revolving in some fixed orbits in space is a Celestial Body.
- Some celestial bodies are very big, hot and are made up of gases.
- Examples of celestial bodies are the Sun, the Moon, the Planets and all the objects shining in the night sky.

Celestial bodies that have their own heat and light, which they emit in large amounts are called stars. The Sun is a star, which has its own source of energy.

SUPERNOVA

- A Supernova is the explosion of a star.
- It is the largest explosion that takes place in space.

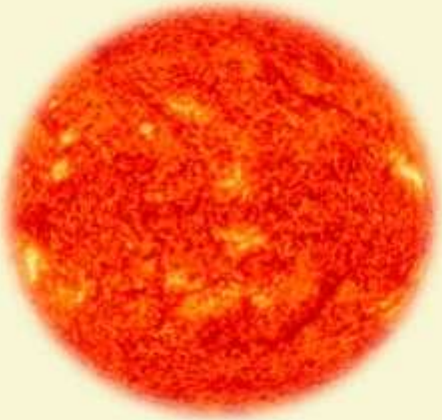
BLACK HOLE



- A **black hole** is a place in space where gravity pulls so much that even light cannot get out.
- The gravity is so strong because matter has been squeezed into a tiny space.
- This happens, when a star is dying.
- A black hole cannot be seen because strong gravity pulls off all the light into the middle of the black hole.

Learning Space

FACTS ABOUT SUN

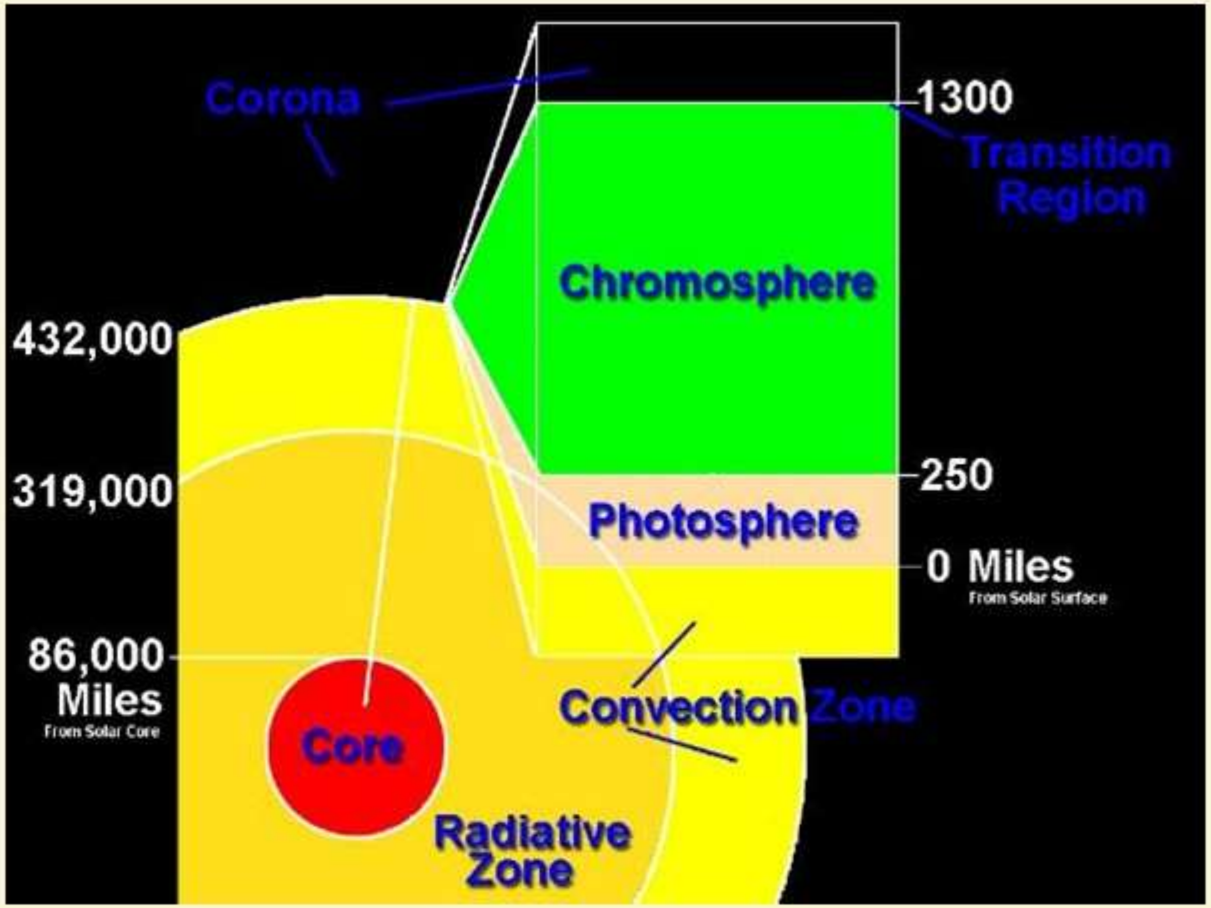


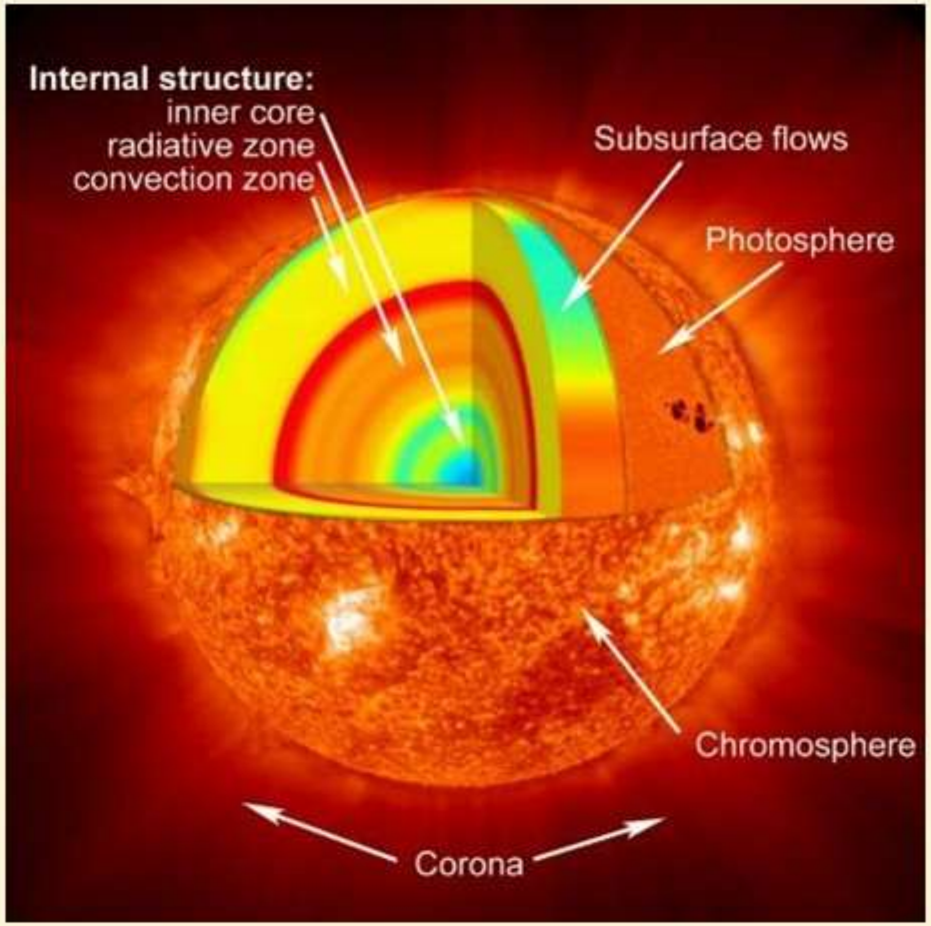
The Sun being a star generates light and heat energy via nuclear fusion process taking inside its core.

1. Sun is at the centre of our solar system.
2. It makes up 99.8% of the mass of the entire solar system.
3. Sun is a star and a star does not have a solid surface. But, it is a ball of gas (92.1% Hydrogen and 7.8% Helium) held together by its own gravity.
4. Solar atmosphere is where we see features such as sunspots and solar flares on the Sun.
5. The temperature at the Sun's core is about 27 million^{0}F (15 million^{0}C)

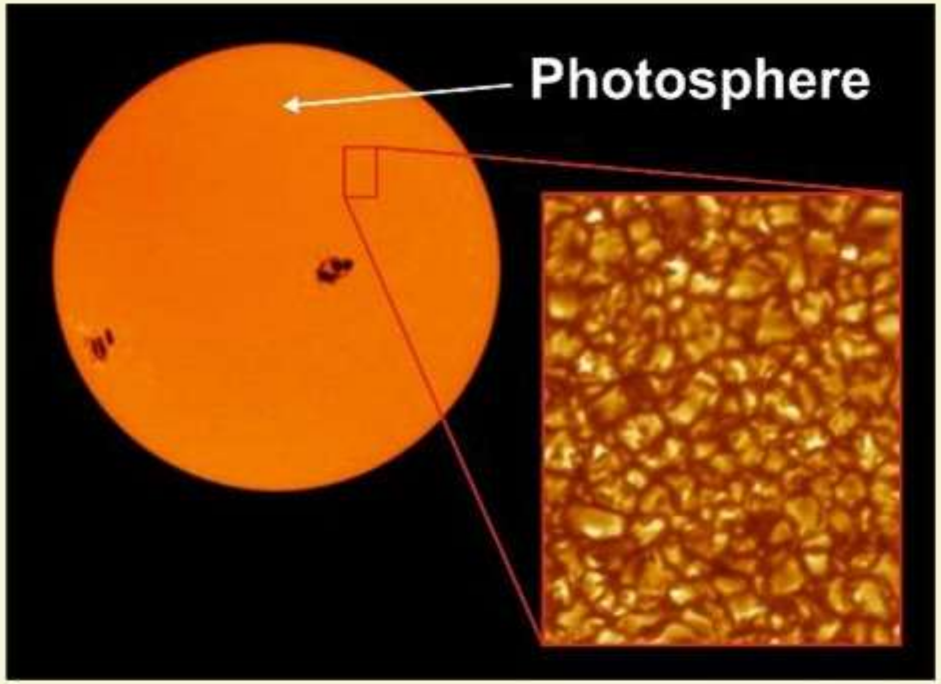
MORE ON SUN

1. Deep in the Sun's core, nuclear fusion reaction takes place which converts Hydrogen to Helium and this process generates energy.
2. Particles of light called photons carry this energy.
3. The Sun's surface or atmosphere is divided into three regions i.e. the Photosphere, the Chromosphere and the Solar Corona.
4. The Photosphere is the visible surface of the Sun and the lowest layer of the atmosphere.
5. Just above the Photosphere are Chromosphere and Corona and these are seen only during a Solar Eclipse.
6. In addition to light, the Sun radiates heat and a steady stream of charged particles known as solar wind.
7. This wind blows about 450 km/s through out the solar system.


Corona
1300
Transition Region
Chromosphere
432,000
250
319,000
Photosphere
0 Miles
From Solar Surface
86,000
Miles
From Solar Core
Convection Zone
Core
Radiative Zone


Internal structure:
inner core
radiative zone
convection zone
Subsurface flows
Photosphere
Chromosphere
Corona

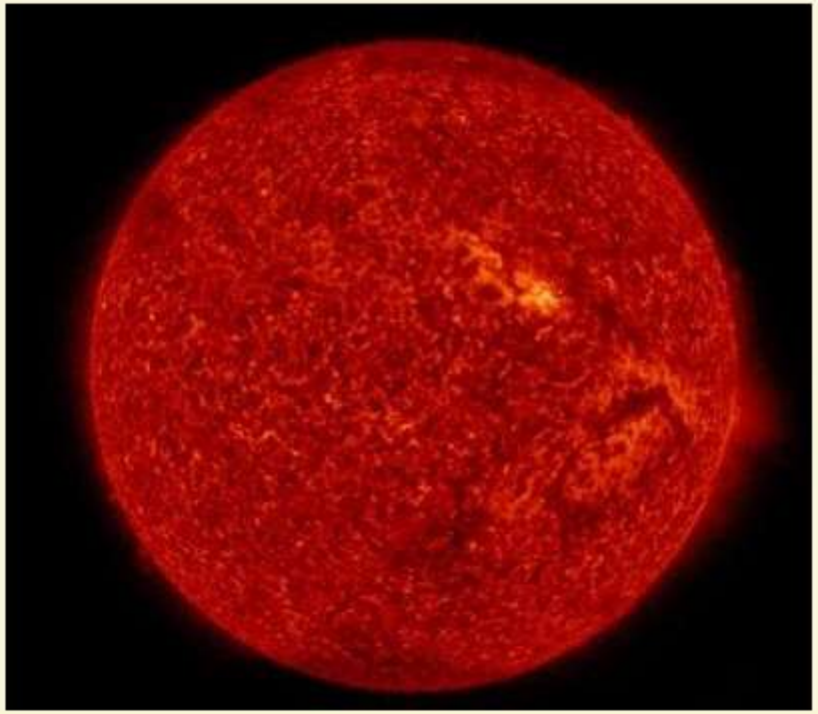
PHOTOSPHERE



Sunspots emerge, when the sun's magnetic field breaks through the surface.

1. It is the lowest layer of the Sun's atmosphere.
2. It is about 400 kms thick.
3. This is the layer, where the Sun's energy is released as light.
4. The Photosphere is also the source of solar flares.
5. These are tongues of fire that extend hundreds of thousands of miles above the Sun's surface.
6. Solar flares produce burst of x-rays, ultraviolet radiation, electromagnetic radiation and radio waves.

CHROMOSPHERE



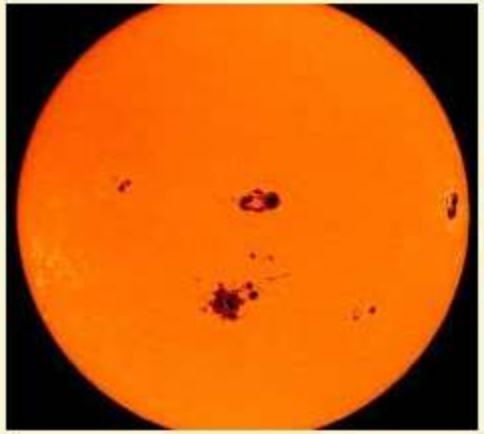
1. It emits reddish glow as super-heated hydrogen burns off.
2. It can be seen only during Solar Eclipse.
3. The Chromosphere play a role in conducting heat from the interior of the Sun to the outermost layer i.e. the Corona.

CORONA

1. The third layer is Corona.
2. It can only be seen during total Solar Eclipse.
3. Temperature in the Sun's Corona can go to as high as 2 million^{0}C.

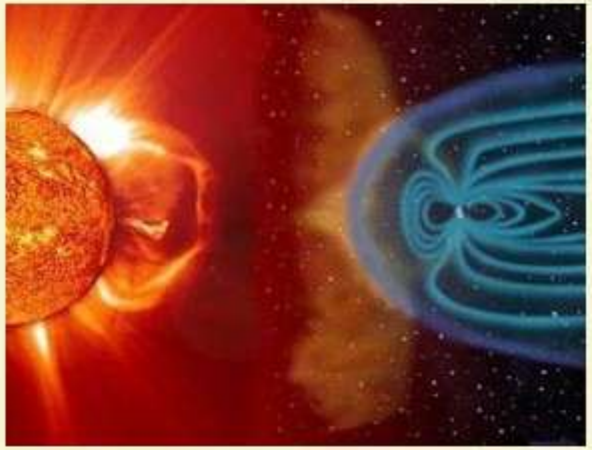
Why the Corona is up to 300 times hotter than Photosphere, despite being farther from the solar core has remained a long time mystery.

SUNSPOT



- Dark areas on the solar surface containing strong magnetic fields that are constantly shifting.
- They occur when the strong magnetic fields emerge through the solar surface.

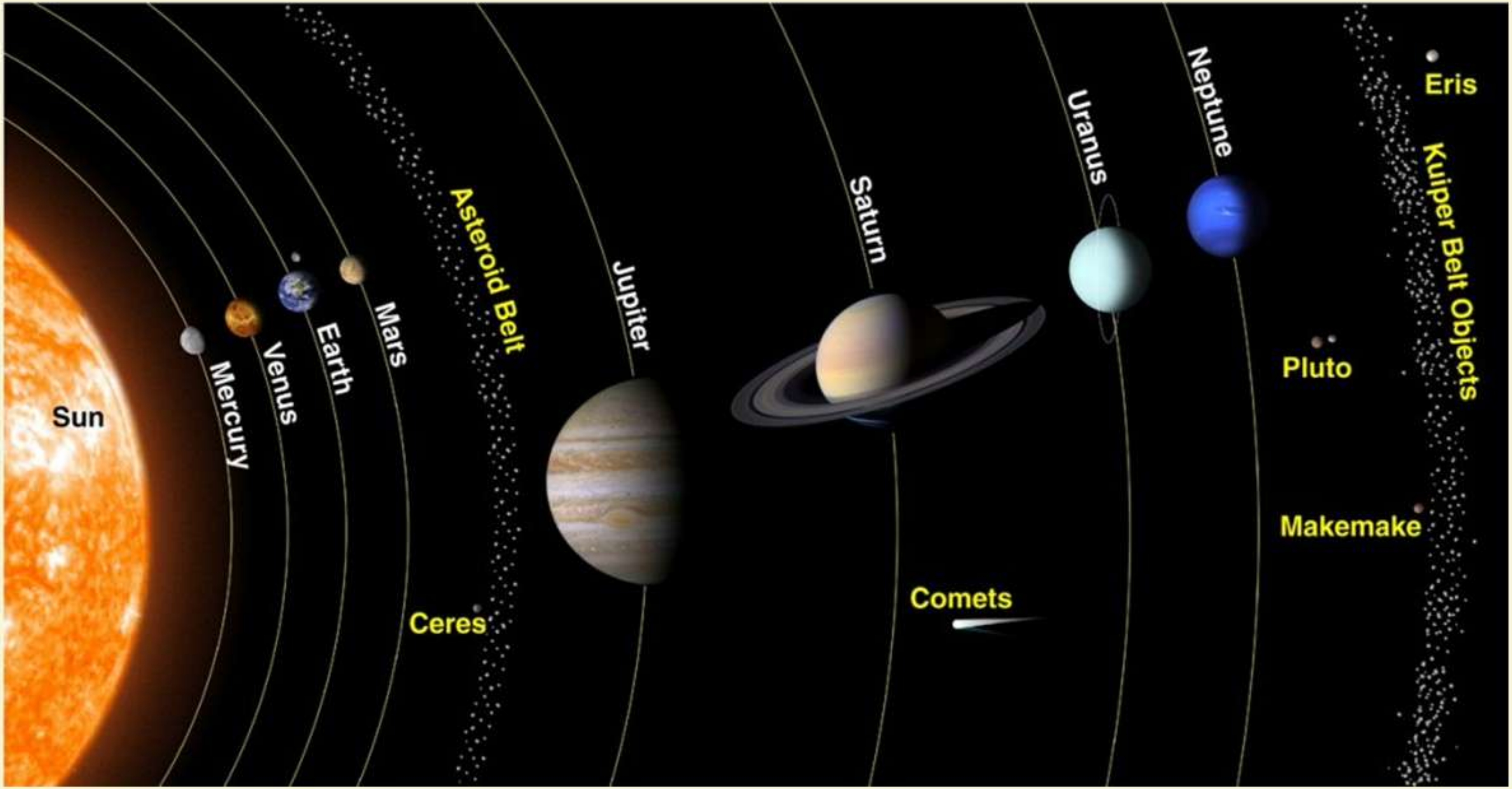
SOLAR WIND



- Stream of charged particles released from the upper atmosphere of the Sun called the Corona.
- The temperature of the Corona is so high, that the Sun's gravity cannot hold on to it.

SOLAR FLARE

- It is intense burst of radiation coming from the release of magnetic energy associated with the sunspots.
- These are our solar system's largest explosive events.
- Flares are also the sites where particles (electrons, protons and heavier particles) are accelerated.


Sun
Mercury
Venus
Earth
Mars
Asteroid Belt
Ceres
Jupiter
Saturn
Comets
Uranus
Neptune
Pluto
Makemake
Eris
Kuiper Belt Objects

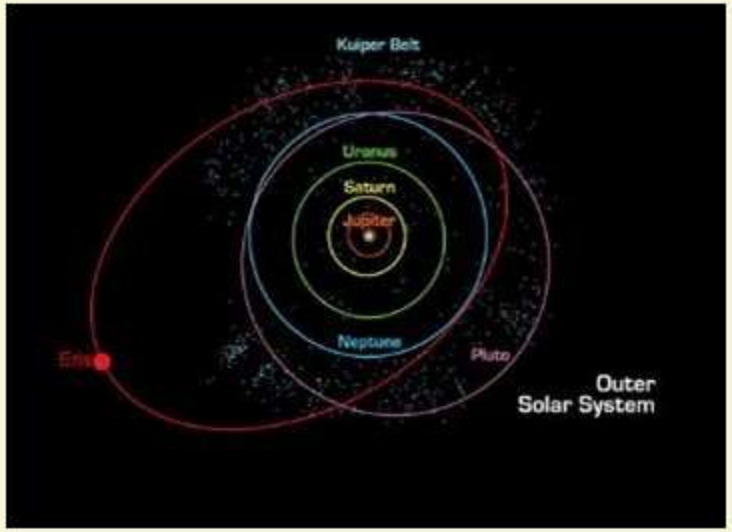
Mercury 8^{th}	Venus 6^{th}	Earth 5^{th}	Mars 7^{th}	Jupiter 1^{st}	Saturn 2^{nd}	Uranus 3^{rd}	Neptune 4^{th}
INNER PLANETS				OUTER PLANETS / JOVIAN PLANETS (99% mass)			
Closely spaced Rocky and Terrestrial Silicates and Metals				Gas Giants Hydrogen & Helium		Ice Giants Water, Ammonia & Methane	
No Moon	No Moon	1 Moon	2 Moons	69 Moons	61 Moons	27 Moons	14 Moons
Smallest	Hottest @462°C			Largest & Heaviest			
	Morning Star/ Evening Star	Supports Life	Red Planet	Has Rings	Has Rings	Has Rings	Has Rings
	Earth's Twin	Blue Planet	Highest Mountain (Olympus Mons)	Lord of the Heavens			
	Brightest Planet						
88 Days	225 Days	365 Days	688 Days	11.9 Years	29.5 Years	84 Years	164.8 Years

Inner and outer planets are separated by Asteroid belt

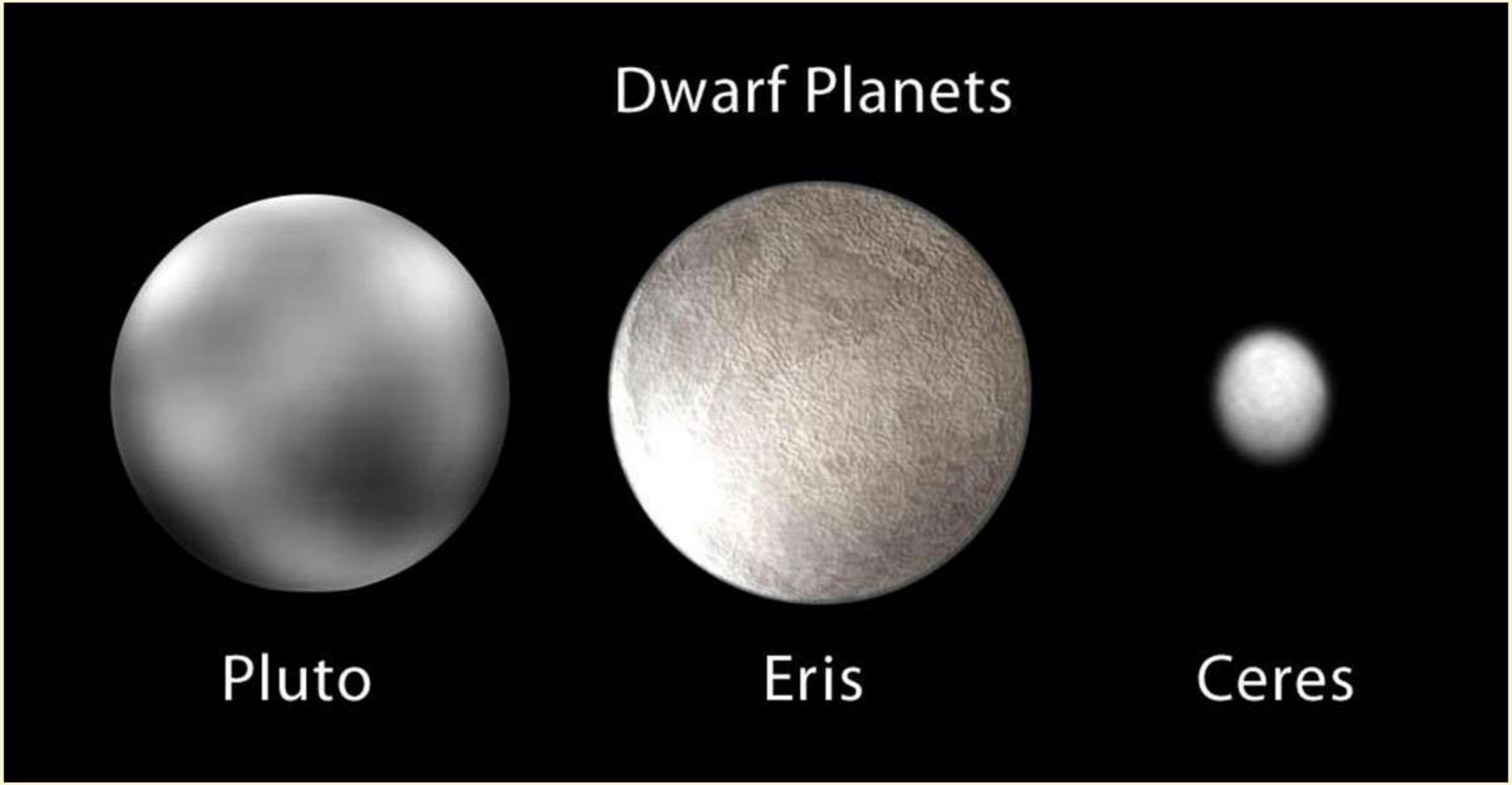
ASTEROIDS

- Apart from the stars, planets and satellites, these are numerous tiny rocky bodies, which move around the Sun and known as **Asteroids**.
- Planetoids or minor planets.
- Leftovers from the formation of our Solar System about 4.6 billion years ago.

DWARF PLANETS

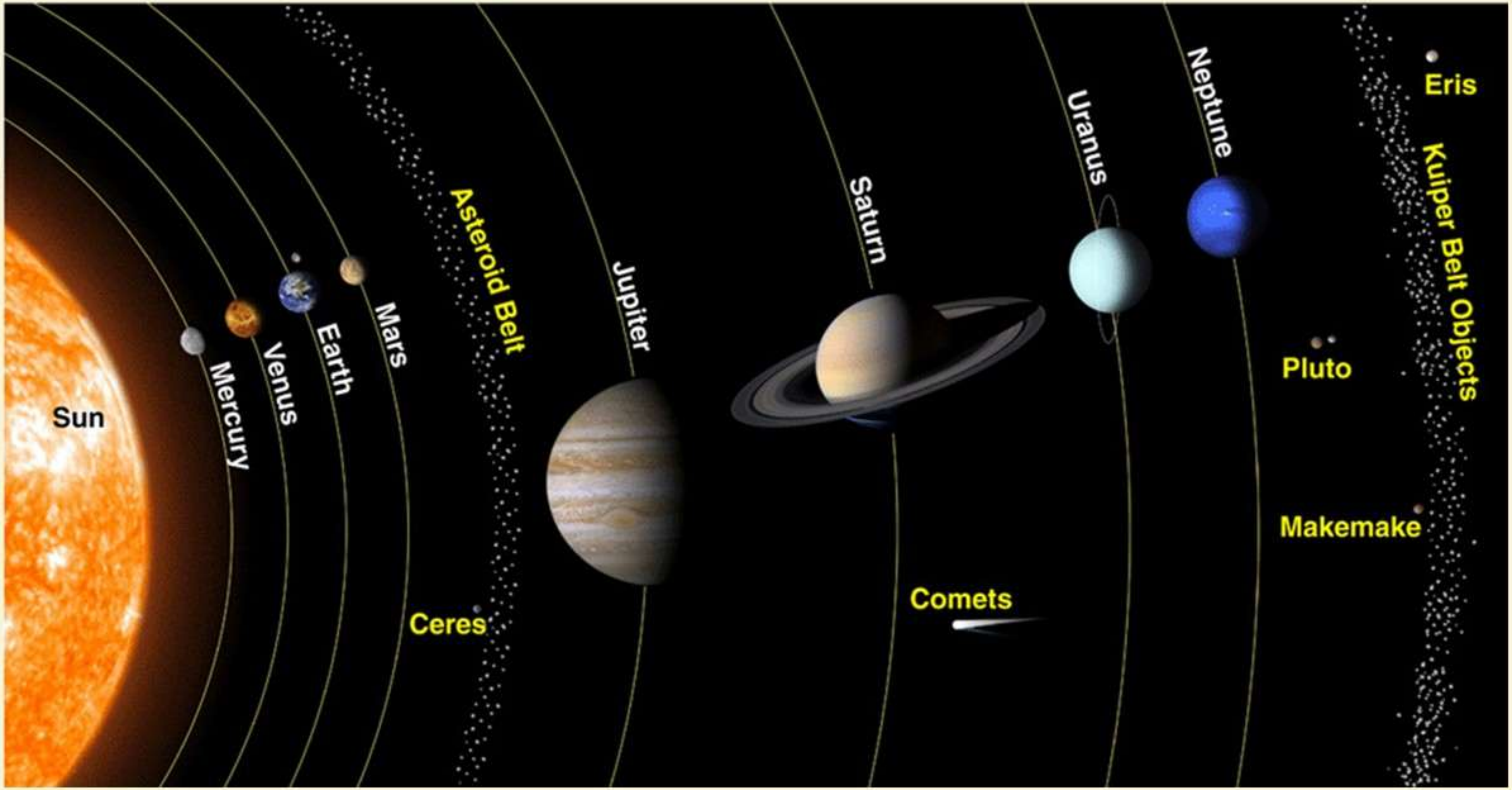


- A **Dwarf Planet** is a planetary-mass object that is neither a planet nor a natural satellite.
- The main distinction between a dwarf planet and a planet is that planets have clear paths around the Sun.
- The meaning is they have not cleared the neighbourhood around their orbits.


Dwarf Planets
Pluto
Eris
Ceres

ASTEROID BELT

- It is the region of space between the orbits of Mars and Jupiter.
- It contains millions of asteroids.
- It is believed that asteroid belt is made up of the material that was never able to form into a planet or of the remains of a planet which broke apart a long time ago.
- The largest asteroid is called Ceres, it is about one quarter the size of our Moon. It is a dwarf planet.
- They are formed probably due to the reason that not all the dust and rocks were pulled together by gravity into planets.


Sun
Mercury
Venus
Earth
Mars
Asteroid Belt
Ceres
Jupiter
Saturn
Comets
Uranus
Neptune
Pluto
Makemake
Eris
Kuiper Belt Objects

COMETS

- Comets are balls of rock and ice that grow tails as they approach the Sun, in the course of their highly elliptical orbits.
- As comets heat up, gas and dust are expelled and trail behind them.
- The Sun illuminates this trail, causing it to glow.
- The glowing trails are visible in the night sky.

Comets tend to have more chemical compounds that vaporize, when heated such as water and have got more elliptical orbits than asteroids.

METEOROIDS, METEORS, METEORITES

- Meteoroid is a small particle from a comet or asteroid orbiting the Sun.
- When it enters the Earth's atmosphere and vaporizes, it results in a light phenomena, which is known as a Shooting Star or Meteor.
- A meteoroid that survives its passage through the Earth's atmosphere and which lands upon the Earth's surface is known as Meteorite.

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For Suggestions:
suggestions@learningspace.in

To Contact us:
info@learningspace.in

OUR TEAM

G. V. Rao

Amrita Naidu
D. Sunil Kumar
L. V. Krishna
D. Chaitanya
K. Srikanth
G. Ravi Babu
D. Joji
K. Victor Babu

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